EKKSC

Superior® 298 Material

EKK Eagle Semicon Components, Inc

www.EKKSC.com

EKKSC Superior® 298 Material

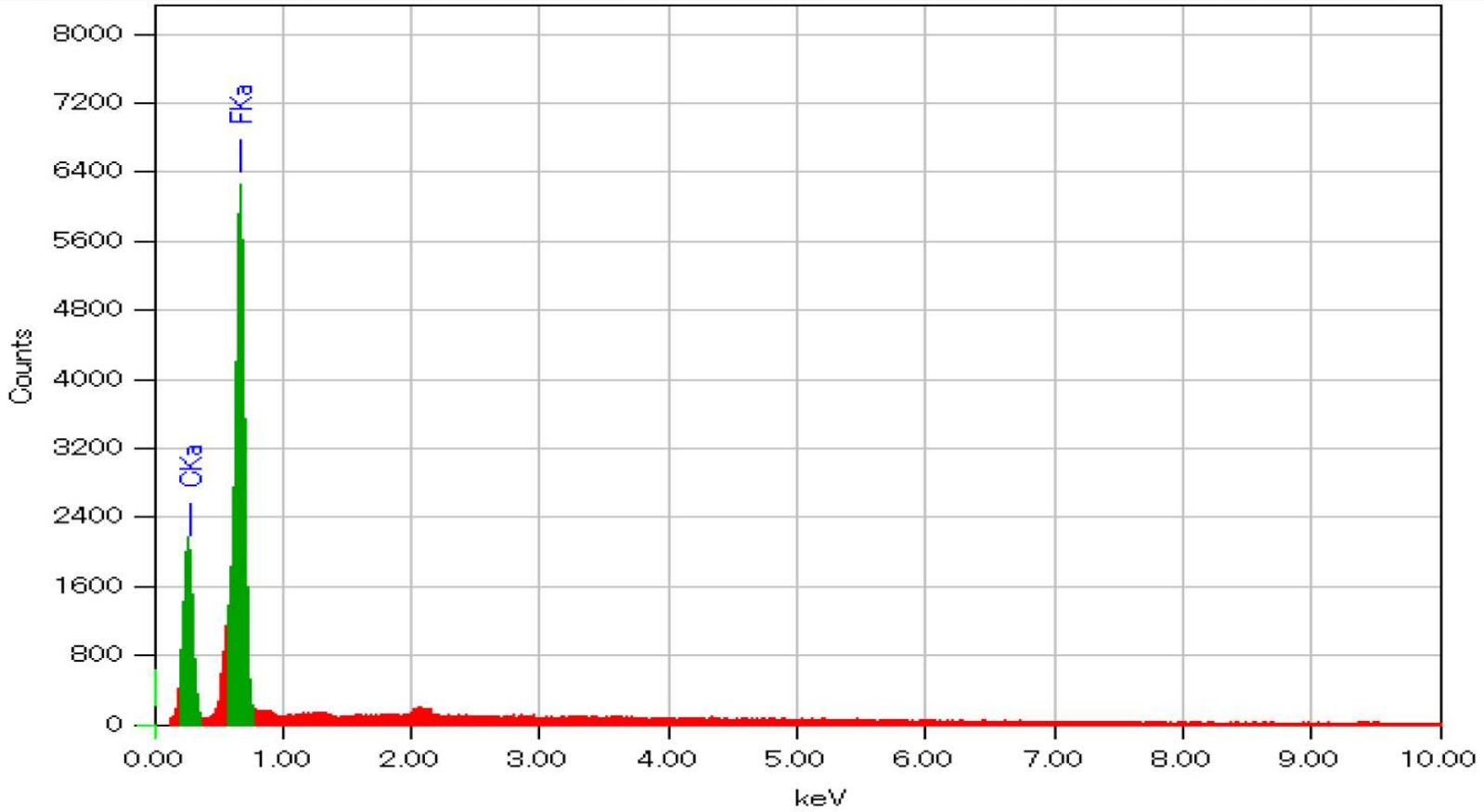
Superior® 298 is specifically designed as a exceptionally clean unfilled FFKM for high temperature plasma applications and has excellent chemical compatibility in wet environments.

- Excellent plasma resistance
- Low particulation and extractables in a wide range of plasma, acid, and solvent environments.
- Unmatched compression set and high temperature performance for an unfilled FFKM.

Physical Properties

Physical Properties	Superior® 298
Material Type	FFKM - Unfilled
Color	Violet
Hardness, Duro-A	67
Tensile Strength, Mpa	16.9
100% Modulus, Mpa	3.4
Ultimate Elongation, %	214
Compression Set, 70 hrs @ 200° C	16.3
Compression Set, 70 hrs @ 250° C	13.1
Compression Set, 70 hrs @ 275° C	17.8
Compression Set, 70 hrs @ 300° C	10.3
Total OutGassing, 250° C, ppm	3.7
Plasma Weight Loss. %, 1800W ICP, 20 hrs, 0.15 Torr, O2	0.39
Adhesion Testing: Stainlesss Steel - 24 hours @ 25% compression and 24 hours rested before load tearing	
24 hrs @ 100°C	154N
25 hrs @ 150°C	307N
26 hrs @ 200°C	341N

ICP-MS for Superior® 298 Material

TM by ICP-MS for Selected Elements in Polymer Material			
Parameter	**RL**	**Units**	**EKKSC Superior® 298**
Aluminum (Al)	5	ppb (ng/g)	280
Barium (Ba)	5	ppb (ng/g)	19
Calcium (Ca)	50	ppb (ng/g)	650
Chromium (Cr)	30	ppb (ng/g)	94
Copper (Cu)	30	ppb (ng/g)	22
Iron (Fe)	50	ppb (ng/g)	550
Lead (Pb)	5	ppb (ng/g)	2
Lithium (Li)	5	ppb (ng/g)	<1
Magnesium (Mg)	5	ppb (ng/g)	27000
Manganese (Mn)	5	ppb (ng/g)	5
Nickel (Ni)	5	ppb (ng/g)	27
Potassium (K)	50	ppb (ng/g)	8400
Sodium (Na)	30	ppb (ng/g)	670
Strontium (Sr)	5	ppb (ng/g)	4
Titanium (Ti)	50	ppb (ng/g)	130
Zinc (Zn)	50	ppb (ng/g)	110

SEM/EDX testing of Superior® 298